

Intro To ML Group C



[introtoml.png](#)

Ez a jegyzet egy 2024.06.13.-án megírt magyar vizsga alapján készült el.

1.

(10p) We trained an autoencoder, so we have two functions:

$f : R^D \rightarrow R^d, g : R^d \rightarrow R^D, d \leq D$ where for most $x \in R^D$

it holds that $g(f(x)) \approx x$ **How can this system be used as a generative artificial intelligence? How can we generate new, unseen outputs?**

▼ Answer

Autoencoders are neural networks trained to reproduce their input data at the output layer. By using a bottleneck layer in the middle, autoencoders can learn a compressed representation of the input data, which can be used for generating new samples. By sampling new points from this latent space and feeding them through the decoder g , it is possible to generate new, unseen outputs.

2.

(15p) What are the four options for updating a particle's components in Simplified Swarm Optimization (SSO)?

▼ Answer

Each particle changes its position based on the following information:

- current position
- current velocity
- distance between current position and pbest (best solution achieved by the particle)
- distance between current position and gbest (best solution achieved by the swarm)

3. (30p) A perceptron with two inputs is learning the OR logical function. During one phase of the learning process, the perceptron has the following weights:

$$w_1 = 0.1, w_2 = -0.3$$

$$\text{Threshold} : \Theta = 0.25$$

Learning rate : $\eta = 0.1$

Show the learning process for one epoch.

▼ Answer

Iteration	Inputs	←	Initial weights	←	Output	←	Error
p	$x_1(p)$	$x_2(p)$	$w_1(p)$	$w_2(p)$	$d(p)$	$y(p)$	$e(p)$
1	0	0	0.1	-0.3	0	0	0
2	0	1	0.1	-0.3	1	0	1
3	1	0	0.1	-0.2	1	0	1
4	1	1	0.2	-0.2	1	0	1

Additional help:

- Desired output: $d(p) = x_1(p) \text{ OPERATOR } x_2(p)$
- Actual output: $y(p) = \text{step}([x_1(p) * w_1(p) + x_2(p) * w_2(p)] - \theta)$
- Step: IF $x \geq 0$ THEN 1 ELSE 0
- Error: $e(p) = d(p) - y(p)$
- New weights: $w_i = w_i + \eta \cdot x_i(p) \cdot e(p)$

4. (15p) In the ant foraging model, how much pheromone (in units) does an ant deposit on a given cell, if the amount of pheromone perceived in the three possible directions is as follows?

LEFT	STRAIGHT	RIGHT
25	15	10

▼ Answer

Ants deposit a **unit of pheromone** at their current location.

- **"A" pheromone** is deposited if the ant is searching for food.
- **"B" pheromone** is deposited if the ant is carrying food.

5. (15p) What are the main steps in training a policy gradient algorithm?

▼ Answer

1. Initialize the agent
2. Run the policy until termination:
 - a. Record all states, actions, rewards
 - b. Decrease probability of actions that resulted in low reward
 - c. Increase probability of actions that resulted in high reward

6. (15p) What is the basic concept of reinforcement learning? Draw an example to demonstrate the training process! What are the main components of reinforcement learning?

▼ Answer

Reinforcement learning involves an agent interacting with an environment to maximize cumulative rewards over time. Key components include:

- **State Space (S):** Configurations of the environment.
- **Action Space (A):** Possible actions the agent can take.
- **Transition Model (T):** Probabilities of moving between states.
- **Reward Function (R):** Immediate feedback.
- **Policy (π):** Strategy for selecting actions.

